

Mathematics - Probability and Statistics

Thomas de Jaeger
thomas.de-jaeger@epita.fr

Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

Discrete Random Variables

Session 8

Discrete Random Variables

Session 8

Objectives:

- Define discrete random variables, PMFs, and CDFs
- Compute probabilities, mean, and variance
- Use key distributions: uniform, binomial, geometric, negative binomial, hypergeometric, Poisson
- Choose appropriate models for scenarios and relate the distributions

What is a random variable?

It is one of the most important concepts in probability!!

The concept of variable is not new for you: algebra, **variable $X=3$**

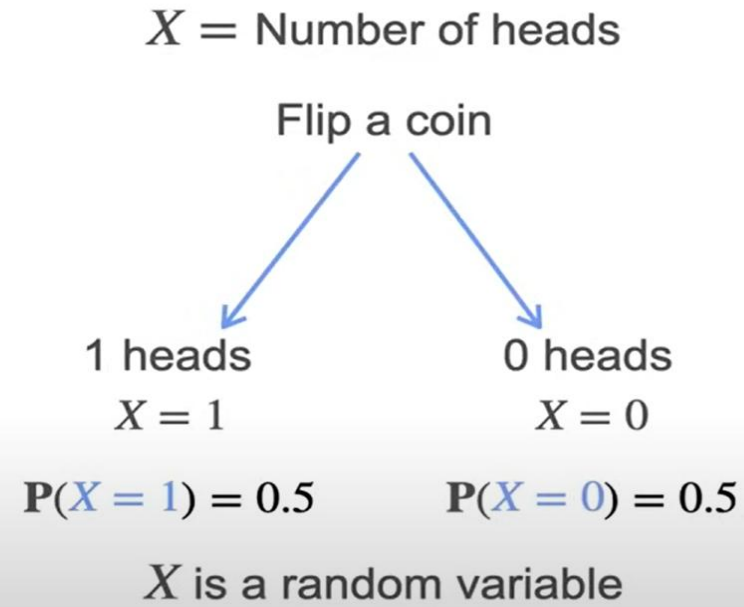
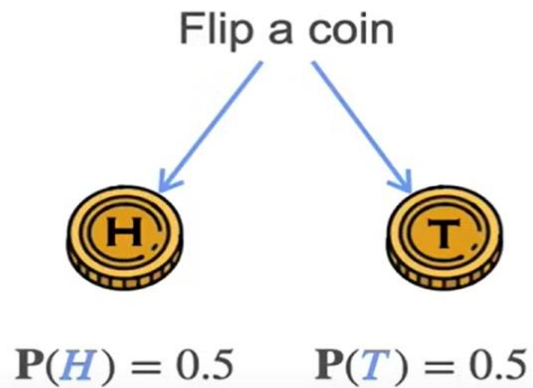
Random variables are different, they can take many values!!

ex: Temperature of the room, the number of head is I toss 10 times,...

What is a random variable?

Definition: Given a random experiment with **sample space Ω** , a **random variable X** is a set function that assigns one and only one real number to each outcome ω

$$X : \Omega \rightarrow \mathbb{R}$$
$$\omega \rightarrow X(\omega)$$



What is a random variable?

Examples:

- Flip a coin ten times. Let $X(\omega)$ be the number of heads in the sequence ω . For example, if $\omega = \text{HHTHHTHHTT}$, then $X(\omega) = 6$.
- You are a football game, Real Madrid –Barcelona. Madrid is playing at home, therefore 80% of the fans are for Madrid.

You select 3 random fans, what is the probability that you have at least 2 Madrid fans?

$\Omega = \{\text{MMM}, \text{MMB}, \text{MBB}, \text{BBB}, \text{BBM}, \text{BMB}, \text{MBM}, \text{BMM}\}$

X = number of Madrid fan

So X can be 0, 1, 2, 3

We need to derive $P(X=2)$ and $P(X=3)$.

$X=2: \{\text{MMB}, \text{MBM}, \text{BMM}\} \rightarrow P(X=2) = 3 \cdot (0.8 \cdot 0.8 \cdot 0.2) = 0.384$

$X=3: \{\text{MMM}\} \rightarrow P(X=3) = (0.8 \cdot 0.8 \cdot 0.8) = 0.512$

$P(X \geq 2) = 0.384 + 0.512 = 0.896$

What is a discrete random variable?

Definition:

A random variable is a **discrete random variable** if:

- there are a finite number of possible outcomes of , or
- there are a countably infinite number of possible outcomes of

Examples:

- Rolling one fair die: $X \in \{1, 2, 3, 4, 5, 6\}$ with $P(X=k)$
- Car passing a sensor in 1 min: Poisson model

Probability mass function

Definition:

The most important way to characterize a random variable is through the probabilities of the values that it can take. For a discrete random variable X , these are captured by the probability mass function (PMF for short) of X , denoted f_X .

If x is any possible value of X , the probability mass of x , denoted $f_X(x)$, is the probability of the event $\{X = x\}$ consisting of all outcomes that give rise to a value of X equal to x : $f_X(x) = P(\{X = x\})$

Examples:

- Rolling one fair die: $X \in \{1, 2, 3, 4, 5, 6\}$ with $P(X=k)=1/6$

Notes:

Upper case characters to denote random variables, and lower case characters to denote real numbers such as the numerical values of a random variable

Probability mass function

Properties: if X must assume one of the values $x_1, x_2, \dots$, then

- All the probabilities are between 0 and 1
 - $0 \leq f_X(x_i) \leq 1$ for $i=0,1,2,\dots,n$
- The sum of all probabilities is 1
 - $\sum f_X(x_i) = 1$
- $f_X(x)=0$ for all other values of x

Probability mass function

Representation: Flip 3 times a coin and count the number of head

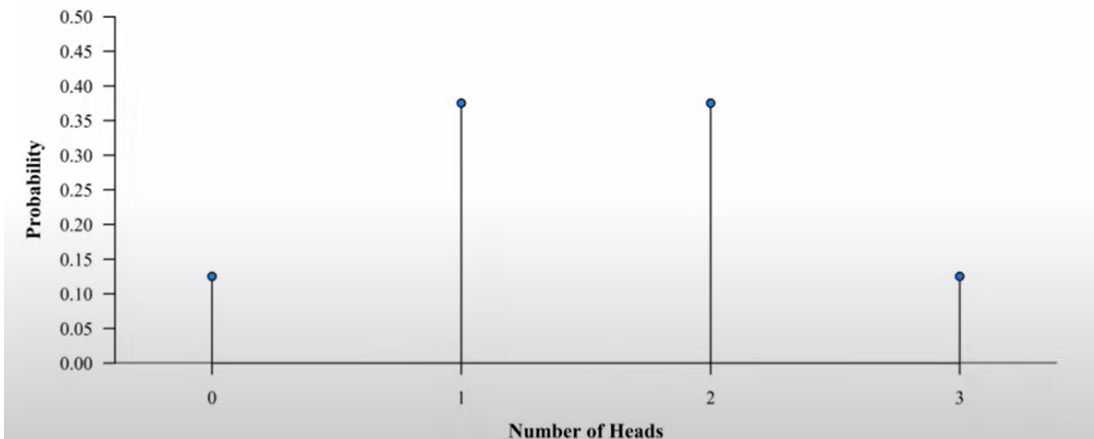
- **Table**

Heads	Flip Outcomes	Probability
0	TTT	$1/8 = 0.125$
1	TTH, THT, HTT	$3/8 = 0.375$
2	HHT, HTH, THH	$3/8 = 0.375$
3	HHH	$1/8 = 0.125$

- **Formula**

$$\mathbb{P}[X = x] = \begin{cases} 0.125 & x = 0 \text{ or } 3 \\ 0.375 & x = 1 \text{ or } 2 \\ 0 & \text{Otherwise} \end{cases}$$

- **Graph**



Exercise

Problem: You have $f(x) = cx^2$ for $x = 1, 2, 3$. Determine the constant c so that the function $f(x)$ satisfies the conditions of being a probability mass function.

Exercise

Solution: You have $f(x) = cx^2$ for $x = 1, 2, 3$. Determine the constant c so that the function $f(x)$ satisfies the conditions of being a probability mass function.

- The sum of all probabilities is 1

$$\sum f(x) = 1$$

$$c \cdot 1^2 + c \cdot 2^2 + c \cdot 3^2 = 1$$

$$14c = 1 \rightarrow c = 1/14$$

Cumulative distribution function (cdf)

Definition:

The cumulative distribution function, or cdf, is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(x) = P(X \leq x).$$

This tells you “what fraction of the time the outcome is at most x .”

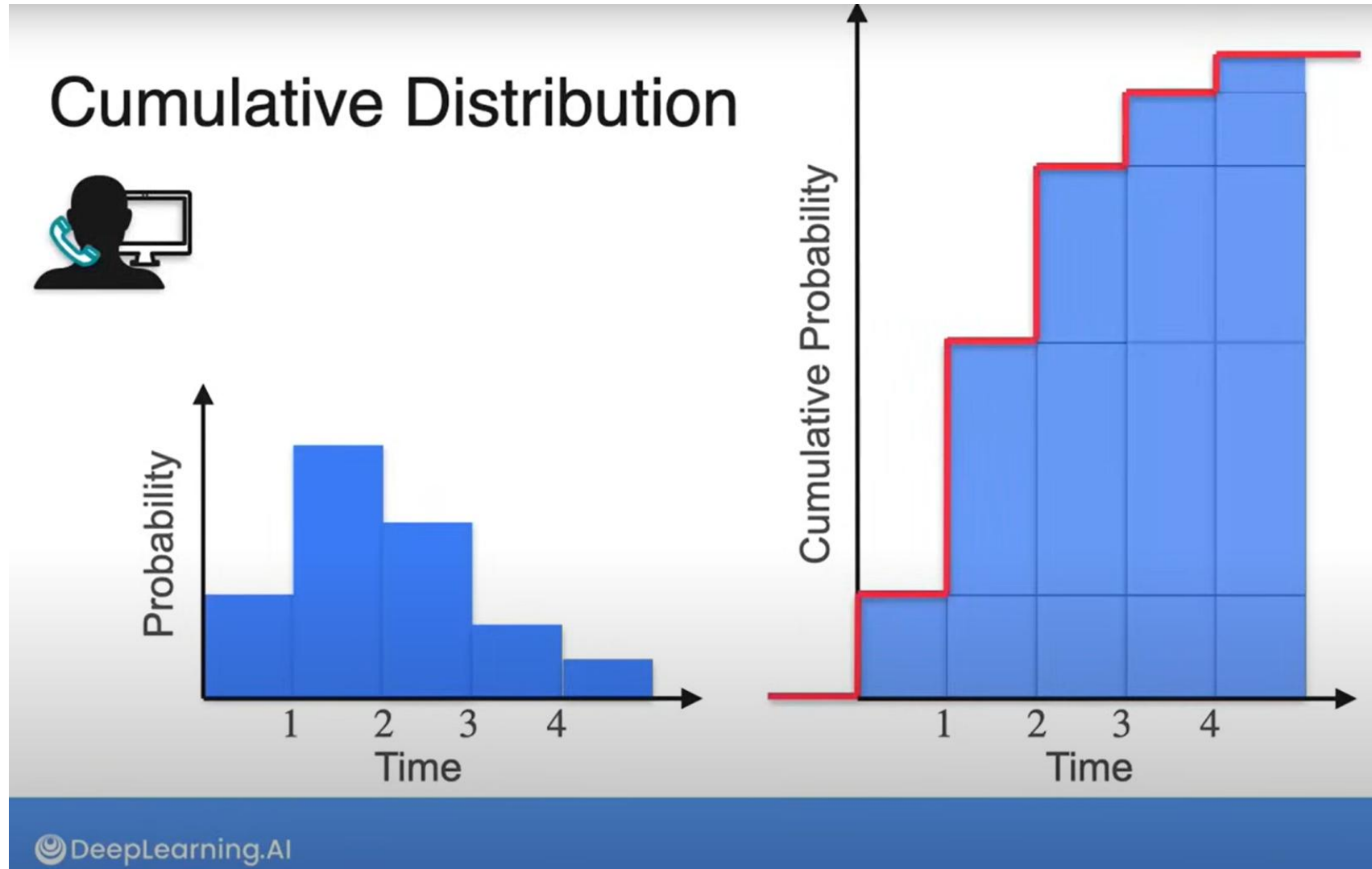
Therefore:

- It never goes down
- It starts near 0 far to the left and approaches 1 far to the right
- For **discrete** variables, it's a **step function**: it jumps at the values where X can actually land
- $P(a < X < b) = F_X(b) - F_X(a)$

Relation between the pmf and cdf:

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

Cumulative distribution function (cdf)



Exercise

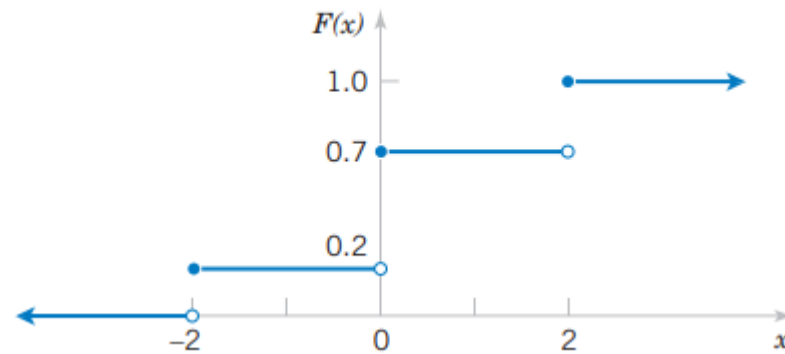
Example: Determine the pmf $f(x)$ for:

$$F(x) = \begin{cases} 0 & x < -2 \\ 0.2 & -2 \leq x < 0 \\ 0.7 & 0 \leq x < 2 \\ 1 & 2 \leq x \end{cases}$$

Exercise

Solution: First represent $F(X)$:

$$F(x) = \begin{cases} 0 & x < -2 \\ 0.2 & -2 \leq x < 0 \\ 0.7 & 0 \leq x < 2 \\ 1 & 2 \leq x \end{cases}$$



From the plot, the only points that receive nonzero probability are -2, 0, and 2

$$f(-2) = 0.2 - 0 = 0.2$$

$$f(0) = 0.7 - 0.2 = 0.5$$

$$f(2) = 1 - 0.7 = 0.3$$

Exercise

Example: A manufacture produce 850 coins. 50 does not have the quality required. Two coins are selected at random, without replacement. Let the random variable X equal the number of nonconforming coin in the sample. What is the cumulative distribution function of X ?

Exercise

Solution:

- **First, derive the pmf. $X=0,1,2$**

$$P(X=0)=800/850*799/849=0.886$$

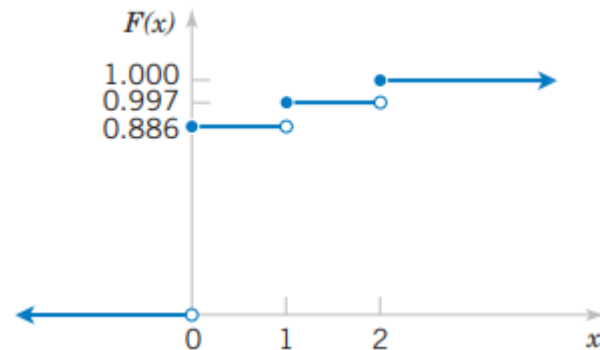
$$P(X=1)=800/850*50/849+50/850*800/849=0.111$$

$$P(X=2)=50/850*49/849=0.003$$

$$f(0)=P(X<0)=0.886$$

$$f(1)=P(X<1)=0.886+0.111=0.997$$

$$f(2)=P(X<2)=0.886+0.111+0.003=1$$



Exercise

Example:

$$F(x) = \begin{cases} 0 & x < -10 \\ 0.25 & -10 \leq x < 30 \\ 0.75 & 30 \leq x < 50 \\ 1 & 50 \leq x \end{cases}$$

(a) $P(X \leq 50)$

(b) $P(X \leq 40)$

(c) $P(40 \leq X \leq 60)$

(d) $P(X < 0)$

(e) $P(0 \leq X < 10)$

(f) $P(-10 < X < 10)$

Exercise

Solution:

- a) $P(X \leq 50) = 0.25 + 0.75 = 1$
- b) $P(X = 40) = 0.75$
- c) $P(40 \leq X \leq 60) = 1 - 0.75 = 0.25$
- d) $P(X < 0) = 0.25$
- e) $P(0 \leq X < 10) = 0$
- f) $P(-10 < X < 10) = 0$

$$F(x) = \begin{cases} 0 & x < -10 \\ 0.25 & -10 \leq x < 30 \\ 0.75 & 30 \leq x < 50 \\ 1 & 50 \leq x \end{cases}$$

Mean and variance

The **mean** or **expected value** of the discrete random variable X , denoted as μ or $E(X)$, is

$$\mu = E(X) = \sum_x xf(x) \quad (3-3)$$

The **variance** of X , denoted as σ^2 or $V(X)$, is

$$\sigma^2 = V(X) = E(X - \mu)^2 = \sum_x (x - \mu)^2 f(x) = \sum_x x^2 f(x) - \mu^2$$

The **standard deviation** of X is $\sigma = \sqrt{\sigma^2}$.

What does that mean?

The mean of a discrete random variable X is a weighted average of the possible values of X , with weights equal to the probabilities.

Mean and variance

Example: Consider two independent coin tosses, each with a $3/4$ probability of a head, and let X be the number of heads obtained. What is the mean value and the variance?

Mean and variance

Solution: Consider two independent coin tosses, each with a $3/4$ probability of a head, and let X be the number of heads obtained. What is the mean value and the variance?

X = number of head

$$X=0 \rightarrow P(X=0)=1/4 \cdot 1/4=1/16$$

$$X=1 \rightarrow P(X=1)=1/4 \cdot 3/4 + 3/4 \cdot 1/4=6/16$$

$$X=2 \rightarrow P(X=2)=3/4 \cdot 3/4=9/16$$

$$E[X]=\sum x f(x)=0 \cdot 1/16+1 \cdot 6/16+2 \cdot 9/16=6/16+18/16=24/16=\mathbf{1.5}$$

$$V(X)=\sum x^2 f(x)-E^2[X]=0^2 \cdot 1/16+1^2 \cdot 6/16+2^2 \cdot 9/16-1.5^2=0.375$$

Mean and variance

Example:

The distribution of the lengths of actual patent lives for new drugs is as follows:

Years, y	3	4	5	6	7	8	9	10	11	12	13
$f(y)$	0.03	0.05	0.07	0.10	0.14	0.20	0.18	0.12	0.07	0.03	0.01

What is the mean patent life for a new drug?

Mean and variance

Solution:

The distribution of the lengths of actual patent lives for new drugs is as follows:

Years, y	3	4	5	6	7	8	9	10	11	12	13
$f(y)$	0.03	0.05	0.07	0.10	0.14	0.20	0.18	0.12	0.07	0.03	0.01

What is the mean patent life for a new drug?

$$\mu_Y = E[Y] = \sum_{y=3}^{13} y f(y) = 3(0.03) + 4(0.05) + \cdots + 12(0.03) + 13(0.01) = 7.9.$$

Mean and variance

Example:

Consider two probability mass functions. The first:

x	3	4	5
f(x)	0.3	0.4	0.3

And, the second:

y	1	2	6	8
f(y)	0.4	0.1	0.3	0.2

Derive the mean.

Do a graph of both pmf. What do you see?

Derive variance for X and Y and verify.

Mean and variance

Solution:

x	3	4	5
f(x)	0.3	0.4	0.3

y	1	2	6	8
f(y)	0.4	0.1	0.3	0.2

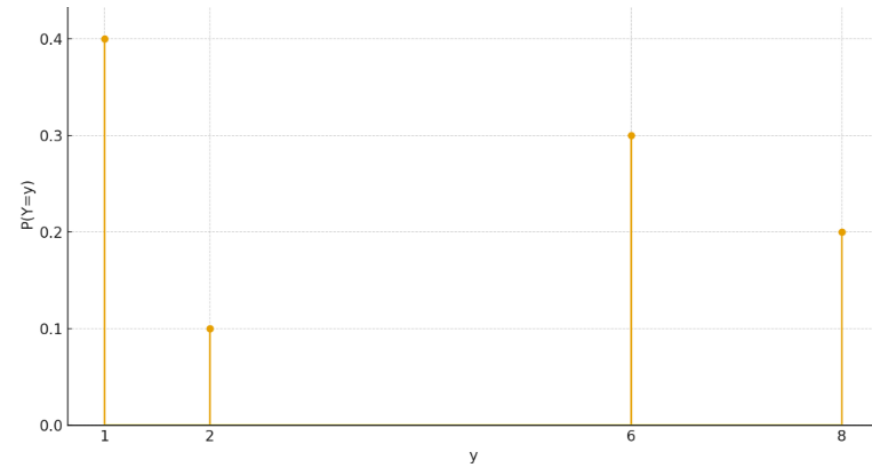
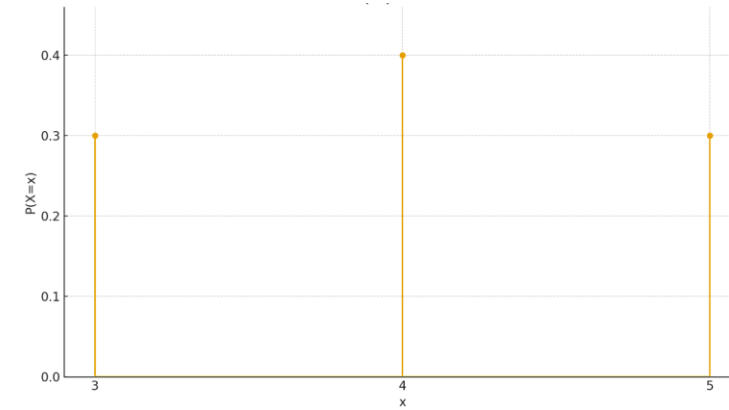
Derive the mean and the variance for X and Y

$$E[X] = 3 \cdot 0.3 + 4 \cdot 0.4 + 5 \cdot 0.3 = 4$$

$$E[Y] = 1 \cdot 0.4 + 2 \cdot 0.1 + 6 \cdot 0.3 + 8 \cdot 0.2 = 4$$

$$V[X] = 3^2 \cdot 0.3 + 4^2 \cdot 0.4 + 5^2 \cdot 0.3 - 4^2 = 0.6$$

$$V[Y] = 1^2 \cdot 0.4 + 2^2 \cdot 0.1 + 6^2 \cdot 0.3 + 8^2 \cdot 0.2 - 4^2 = 8.4$$



Mean and variance

Exercise:

Two new product are to be compared on the basis of revenue potential. Revenue from design A can be predicted quite accurately to be \$3 million.

There is a probability of 0.3 that the revenue from design B will be \$7 million, but there is a 0.7 probability that the revenue will be only \$2 million.

Which design do you prefer?

Mean and variance

Solution:

Two new product are to be compared on the basis of revenue potential. Revenue from design A can be predicted quite accurately to be \$3 million.

There is a probability of 0.3 that the revenue from design B will be \$7 million, but there is a 0.7 probability that the revenue will be only \$2 million.

X=revenue from product A

Y=revenue from product B

$$E[Y] = 0.3 \cdot 7 + 0.7 \cdot 2 = \$3.5 \text{ million}$$

$$\text{Var}[Y] = 0.3 \cdot 7^2 + 0.7 \cdot 2^2 - 3.5^2 = \$5.25 \text{ million} \rightarrow \text{STD}[Y] = \sqrt{5.25} = \$2.3 \text{ million}$$

Important distribution: Uniform

The simplest discrete random variable is one that assumes only a finite number of possible values, each with equal probability.

Definition:

A random variable X has a discrete uniform distribution if each of the n values in its range, say, $x_1, x_2, \dots, x_n$ has equal probability.

$$\text{Then, } f(x_i) = 1/n$$

Mean and variance:

Suppose X is a discrete uniform random variable on the consecutive integers $a, a+1, \dots, b$ for $a \leq b$.

The mean of X is: $E[X] = (a+b)/2$

The variance of X is: $V[X] = [(b-a+1)^2 - 1]/12$

Ex: Die; $a=1, b=6 \rightarrow E[X] = 7/2 = 3.5$ and $V[x] = [(6-1+1)^2 - 1]/12 = 35/12 = 2.9$

Important distribution: Uniform

Exercise: Demonstrate

Suppose X is a discrete uniform random variable on the consecutive integers $a, a+1, \dots, b$ for $a \leq b$.

The mean of X is: **$E[X] = (a+b)/2$**

The variance of X is: **$V[X] = [(b-a+1)^2 - 1]/12$**

Important distribution: Uniform

Solution: Demonstrate

Suppose X is a discrete uniform random variable on the consecutive integers $a, a+1, \dots, b$ for $a \leq b$.

The mean of X is: **$E[X] = (a+b)/2$**

The variance of X is: **$V[X] = [(b-a+1)^2 - 1]/12$**

You have $b-a+1$ values.

Therefore, each value has a probability of $1/(b-a+1)$

$$E[X] = \sum x f(x) = \sum_a^b k/(b-a+1)$$

We know
$$\sum_{k=a}^b k = \frac{(a+b)(b-a+1)}{2}$$

Therefore, $E[X] = (a+b)/2$

$$V[X] = \sum x^2 f(x) = \sum_a^b k^2/(b-a+1)$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Important distribution: Bernoulli

A random variable with only two possible outcomes (e.g., $\{0,1\}$) follows a **Bernoulli distribution** with success probability parameter, p

Example:

- Head on a coin flip. $p=0.5$
- Making a 1 in with a rolling dice: $p=1/6$

Therefore, you can write the probability mass function as:

$$\mathbb{P}[X = x] = \begin{cases} p & x = 1 \\ 1 - p & x = 0 \\ 0 & \text{Otherwise} \end{cases}$$

This can also be written as

$$\mathbb{P}[X = x] = p^x (1 - p)^{1-x}$$

$$E[X] = p \cdot 1 + (1-p) \cdot 0$$

$$\mathbf{E[X]=p}$$

$$\text{Var}[X] = p \cdot 1 + (1-p) \cdot 0^2 - p^2$$

$$\mathbf{Var[X]=p(1-p)}$$

Important distribution: Binomial

Definition: A Binomial random variable is the sum of n independent and identically distributed Bernoulli random variables.

Which is equivalent to say:

A random experiment consists of n **Bernoulli trials** such that

- The trials are independent.
- Each trial results in only two possible outcomes, labeled as “success” and “failure.”
- The probability of a success in each trial, denoted as p , remains constant.

The random variable X that equals the number of trials that result in a success has a **binomial random variable** and a probability mass function of X defined by

$$P_X(k) = \begin{cases} \binom{n}{k} p^k (1-p)^{n-k} & \text{for } k = 0, 1, 2, \dots, n \\ 0 & \text{otherwise} \end{cases}$$

If X is a binomial random variable with parameters p and n ,

$$\mu = E(X) = np \quad \text{and} \quad \sigma^2 = V(X) = np(1-p)$$

Important distribution: Binomial

Exercise: Let X =number of heads in $n=10$ flips. The coin is fair, therefore $p=0.5$

- What is the probability of getting three heads?
- Three tails?
- What is the probability of getting at most three head or at least seven heads?

Important distribution: Binomial

Solution: Let X =number of heads in $n=10$ flips. The coin is fair, therefore $p=0.5$

- A) What is the probability of getting three heads?
- B) Three tails?
- C) What is the probability of getting at most three head or at least seven heads?

A)

$$\mathbb{P}[X = x] = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$\begin{aligned}\mathbb{P}[X = 3] &= \binom{10}{3} 0.500^3 (1 - 0.500)^{10-3} \\ &= (120)(0.125)(0.0078125) \\ &= 0.1171875\end{aligned}$$

B)

$$\mathbb{P}[X = x] = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$\begin{aligned}\mathbb{P}[X = 7] &= \binom{10}{7} 0.500^3 (1 - 0.500)^{10-3} \\ &= (120)(0.125)(0.0078125) \\ &= 0.1171875\end{aligned}$$

C)

$$\mathbb{P}[X \leq 3 \cup X \geq 7] = \mathbb{P}[X \leq 3] + \mathbb{P}[X \geq 7]$$

$$\begin{aligned}&= \mathbb{P}[X = 0] + \mathbb{P}[X = 1] + \mathbb{P}[X = 2] + \mathbb{P}[X = 3] + \\ &\quad \mathbb{P}[X = 7] + \mathbb{P}[X = 8] + \mathbb{P}[X = 9] + \mathbb{P}[X = 10]\end{aligned}$$

Important distribution: Geometric

Assume Bernoulli trials — that is, (1) there are two possible outcomes, (2) the trials are independent, and (3) p , the probability of success, remains the same from trial to trial. Let X denote the number of trials until the first success. Then, the probability mass function of X is:

$$f(x) = P(X = x) = (1 - p)^{x-1}p$$

for $x = 1, 2, \dots$. In this case, we say that X follows a **geometric distribution**.

Important distribution: Geometric

If X is a geometric random variable with parameter p ,

$$\mu = E(X) = 1/p \quad \text{and} \quad \sigma^2 = V(X) = (1 - p)/p^2$$

Demo:

$$\mathbb{E}[X] = \sum_{x=1}^{\infty} x p (1 - p)^{x-1} = p \sum_{x=1}^{\infty} x r^{x-1}$$

$$g'(r) = \sum_{k=1}^{\infty} a k r^{k-1} = 0 + a + 2ar + 3ar^2 + \dots = \frac{a}{(1 - r)^2} = a(1 - r)^{-2}$$

$$\mathbb{E}[X] = p \cdot \frac{1}{(1 - (1 - p))^2} = p \cdot \frac{1}{p^2} = \frac{1}{p}$$

Important distribution: Geometric

Exercise: You select randomly people on a random street in Madrid until you find a person who attended the last home football game. Let p , the probability that he succeeds in finding such a person, equal 0.20. And, let X denote the number of people you select until you find his first success.

What is the probability that you must select 4 people before you find one who attended the last home football game?

Now, how many people should we expect you need to select before you find one who attended the last home football game?

Important distribution: Geometric

Solution: You select randomly people on a random street in Madrid until you find a person who attended the last home football game. Let p , the probability that he succeeds in finding such a person, equal 0.20. And, let X denote the number of people you select until you find his first success.

What is the probability that you must select 4 people before you find one who attended the last home football game?

$$P[X=4]=0.8*0.8*0.8*0.2=0.1024$$

Now, how many people should we expect you need to select before you find one who attended the last home football game?

$$E[X]=1/p=1/0.2=5$$

$$\text{Var}[X]=(1-p)/p^2=0.8/0.2^2=20$$

Important distribution: Negative Binomial

A generalization of a geometric distribution in which the random variable is the number of Bernoulli trials required to obtain r successes results in the **negative binomial distribution**.

In a series of Bernoulli trials (independent trials with constant probability p of a success), let the random variable X denote the number of trials until r successes occur. Then X is a **negative binomial random variable** with parameters $0 < p < 1$ and $r = 1, 2, 3, \dots$, and

$$f(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r \quad x = r, r+1, r+2, \dots$$

If X is a negative binomial random variable with parameters p and r ,

$$\mu = E(X) = r/p \quad \text{and} \quad \sigma^2 = V(X) = r(1-p)/p^2$$

Important distribution: Negative Binomial

Example: A Web site contains **three identical computer servers**. Only one is used to operate the site, and the other two are spares that can be activated in case the primary system fails. The probability of a failure in the primary computer (or any activated spare system) from a request for service is **0.0005**.

Assuming that each request represents an independent trial, what is the mean number of requests until failure of all three servers?

What is the probability that all three servers fail within five requests?

Important distribution: Negative Binomial

Solution:

Here, we have negative binomial distribution with $p=0.0005$ and $r=3$.

- Therefore, $E[X]=3/0.0005=6000$ requests.
- $P[X \leq 5] = P[X=3] + P[X=4] + P[X=5]$ ($P[X=2]=0$ because you need at least 3 requests for all the servers to fail).

$$f(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r$$

$$\begin{aligned} P(X \leq 5) &= P(X=3) + P(X=4) + P(X=5) \\ &= 0.0005^3 + \binom{3}{2} 0.0005^3 (0.9995) \\ &\quad + \binom{4}{2} 0.0005^3 (0.9995)^2 \\ &= 1.25 \times 10^{-10} + 3.75 \times 10^{-10} + 7.49 \times 10^{-10} \\ &= 1.249 \times 10^{-9} \end{aligned}$$

Important distribution:Poisson

The Poisson distribution is one of the most widely used probability distributions. It is usually used in scenarios where we are counting the occurrences of certain events in an interval of time or space.

Definition: Let the discrete random variable X denote the number of times an event occurs in an interval of time (or space). Then X may be a Poisson random variable with $x=0,1,2,\dots$

Ex: Number of photons during 1h observation
Number of cars in 2h on the périphérique

Definition:

If X is a Poisson random variable, then the probability mass function is:

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

for $x = 0, 1, 2, \dots$ and $\lambda > 0$, where λ will be shown later to be both the mean and the variance of X .

Important distribution:Poisson

- $E[X] = \lambda$

Dem: Use $e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$

- $\text{Var}[X] = \lambda$

Important distribution:Poisson

- $E[X] = \lambda$

$$\begin{aligned} E(X) &= \sum_{k=1}^{+\infty} k \mathbb{P}(X = k) \\ &= \sum_{k=1}^{+\infty} k e^{-\lambda} \frac{\lambda^k}{k!} \\ &= e^{-\lambda} \sum_{k=1}^{+\infty} \frac{\lambda^k}{(k-1)!} \\ &= \lambda e^{-\lambda} \sum_{k=1}^{+\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda \end{aligned}$$

- $\text{Var}[X] = \lambda$

$$\begin{aligned} V(X) &= E(X^2) - (E(X))^2 \\ &= \sum_{k=1}^{+\infty} k^2 \mathbb{P}(X = k) - \lambda^2 \\ &= \sum_{k=1}^{+\infty} k^2 e^{-\lambda} \frac{\lambda^k}{k!} - \lambda^2 \\ &= \lambda e^{-\lambda} \sum_{k=1}^{+\infty} \frac{k \lambda^{k-1}}{(k-1)!} - \lambda^2 \\ &= \lambda e^{-\lambda} \sum_{k=1}^{+\infty} \frac{d}{d\lambda} \frac{\lambda^k}{(k-1)!} - \lambda^2 \end{aligned}$$

$$\begin{aligned} &= \lambda e^{-\lambda} \frac{d}{d\lambda} \sum_{k=1}^{+\infty} \frac{\lambda^k}{(k-1)!} - \lambda^2 \\ &= \lambda e^{-\lambda} \frac{d}{d\lambda} \left[\lambda \sum_{k=1}^{+\infty} \frac{\lambda^{k-1}}{(k-1)!} \right] - \lambda^2 \\ &= \lambda e^{-\lambda} \frac{d}{d\lambda} [\lambda e^{\lambda}] - \lambda^2 \\ &= \lambda e^{-\lambda} (\lambda + 1) e^{\lambda} - \lambda^2 \\ &= \lambda (\lambda + 1) - \lambda^2 \\ &= \lambda \end{aligned}$$

Important distribution:Poisson

Exercices: The number of emails that I get in a weekday can be modeled by a Poisson distribution with an average of 0.2 emails per minute.

- What is the probability that I get no emails in an interval of length 5 minutes?
- What is the probability that I get more than 3 emails in an interval of length 10 minutes?

Important distribution:Poisson

Solution: The number of emails that I get in a weekday can be modeled by a Poisson distribution with an average of 0.2 emails per minute.

- What is the probability that I get no emails in an interval of length 5 minutes?

X =number of email in 5min

$E[X]=\lambda=0.2*5=1$ mail in average during 5min

$P(X=0)=e^{-\lambda}=e^{-1}=0.37$

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

- What is the probability that I get more than 3 emails in an interval of length 10 minutes?

X =number of emails in 10 min

$E[X]=\lambda=0.2*10=2$ mails in average during 10min

$P(X>3)=1-P(X\leq 3)$

$=1-(P(X=0)+P(X=1)+P(X=2)+P(X=3))$

$=0.1429$

Important distribution:Poisson

Exercices: Astronomers treat the number of stars in a given volume of space as a Poisson random variable. The density in the Milky Way Galaxy in the vicinity of our solar system is one star per 250 cubic light-years.

- (a) What is the probability of two or more stars in 250 cubic light-years?
- (b) How many cubic light-years of space must be studied so that the probability of one or more stars exceeds 0.95?

Important distribution:Poisson

Solution: Astronomers treat the number of stars in a given volume of space as a Poisson random variable. The density in the Milky Way Galaxy in the vicinity of our solar system is one star per 250 cubic light-years.

(a) What is the probability of two or more stars in 250 cubic light-years?

For $V=250$, you have in average 1 star, therefore $E[X]=1=\lambda$

$P(X \geq 2) = 1 - P(X=0) - P(X=1) = 1 - e^{-1} - e^{-1} = 1 - 2/e = 0.26$

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

(b) How many cubic light-years of space must be studied so that the probability of one or more stars exceeds 0.95?

$P(X \geq 1) = 0.95$

$0.95 = 1 - P(0) = 1 - e^{-\lambda} = 0.95$ therefore, $e^{-\lambda} = 0.05$, $\lambda = 2.99$

Volume should be $2.99 \times 250 = 750$

Important distribution:Poisson

Exercices: Let X equal the number of typos on a printed page with a mean of 3 typos per page.

- What is the probability that a randomly selected page has **at least one typo** on it?
- What is the probability that a randomly selected page has **at most one typo** on it?

Important distribution:Poisson

Solution: Let X equal the number of typos on a printed page with a mean of 3 typos per page.

$$E[X] = \lambda = 3$$

- What is the probability that a randomly selected page has **at least one typo** on it?

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$P(X \geq 1) = 1 - P(X=0) = 1 - e^{-3} = 0.95$$

- What is the probability that a randomly selected page has **at most one typo** on it?

$$P(X < 1) = P(X=0) + P(X=1) = e^{-3} + e^{-3} * 3 = 4 e^{-3} = 0.19$$

Important distribution: Hypergeometric

Definition:

A set of N objects contains

K objects classified as successes

$N - K$ objects classified as failures

A sample of size n objects is selected randomly (without replacement) from the N objects, where $K \leq N$ and $n \leq N$.

Let the random variable X denote the number of successes in the sample. Then X is a **hypergeometric random variable** and

$$f(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}} \quad x = \max\{0, n + K - N\} \text{ to } \min\{K, n\}$$

Examples: Number of hearts draw from a card deck without replacement

Important distribution: Hypergeometric

Definition:

If X is a hypergeometric random variable with parameters N , K , and n , then

$$\mu = E(X) = np \quad \text{and} \quad \sigma^2 = V(X) = np(1 - p) \left(\frac{N - n}{N - 1} \right)$$

where $p = K/N$.

Important distribution: Hypergeometric

Exercise: A batch of parts contains 100 parts from a local supplier of tubing and 200 parts from a supplier of tubing in the next state. If four parts are selected randomly and without replacement, what is the probability they are all from the local supplier?

Important distribution: Hypergeometric

Solution: A batch of parts contains 100 parts from a local supplier of tubing and 200 parts from a supplier of tubing in the next state. If four parts are selected randomly and without replacement, what is the probability they are all from the local supplier?

$$N=100+200=300$$

X=number of parts in the sample from the local supplier

$$K=100$$

$$n=4$$

$$P(X = 4) = \frac{\binom{100}{4} \binom{200}{0}}{\binom{300}{4}} = 0.0119$$

$$f(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

Important distribution: Hypergeometric

Exercise: Let X be the number of spades drawn out of 4 draws from a deck of card without replacement.

What is the probability of getting one spade?

What is the expected number of spades?

Important distribution: Hypergeometric

Solution: Let X be the number of spades drawn out of 4 draws from a deck of card without replacement.

$N=52$

$n=4$

$K=13$

What is the probability of getting one spade?

$x=1$

$$\Pr(X = 1) = \frac{\binom{13}{1} \binom{39}{3}}{\binom{52}{4}} \approx 0.43885$$

What is the expected number of spades?

$$E[X] = nK/N = 4 \cdot 13 / 52 = 1$$

$$f(x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

Summary

Type of distribution	Parameters	Random variables values	Probability	Mean value	Dispersion
Binomial	n, p, (q)	m=0,1..., n	$P_{m,n} = C_n^m p^m q^{n-m}$	np	npq
Geometric	n, p, (q)	m=0,1, 2...	$P_m = pq^{m-1}$	$1/p$	q/p^2
Poisson	a	m=0,1, 2...	$P_m = \frac{a^m}{m!} e^{-a}$	a	a
Hypergeometric	N, p, n, M	m=0,1..., n	$P_{m,n} = \frac{C_M^m C_{N-M}^{n-m}}{C_N^n}$	np	$(npq) \times \left(1 - \frac{n-1}{N-1}\right)$
Note: q=1-p					